

Your grade: 100%

Next item →

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

1. There are a number of possible definitions of a matrix square root. If $\Sigma = \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix}$, then what is $\sqrt{\Sigma}$ in the sense used for the sigma-point method?

1 / 1 point

- ☐ $\begin{bmatrix} 2 & \sqrt{2} \\ \sqrt{2} & \sqrt{2} \end{bmatrix}$.
- ☒ $\begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$.
- ☐ $\begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$.
- ☐ $\begin{bmatrix} 1.897 & 0.632 \\ 0.632 & 1.265 \end{bmatrix}$.

✔

Correct

Yes. This is the correct answer.

2. Suppose that we store the state-prediction sigma points in matrix form as

1 / 1 point

$$\mathcal{X}_k^{x,-} = \begin{bmatrix} 0.5 & 1.1 & 0.7 & 0.4 & 1.0 \\ 0.6 & 0.2 & 0.2 & 0.7 & 0.8 \end{bmatrix}$$

and that $\alpha_k^{(m)} = 0.2$ for all k . We can compute that

$$\hat{x}_k^- = \begin{bmatrix} 0.74 \\ a \end{bmatrix}.$$

What is a ? Enter your answer with two digits to the right of the decimal point.

0.50

✔

Correct

Yes. That is the correct answer.

3. Suppose that we store the state-prediction sigma points in matrix form as

1 / 1 point

$$\mathcal{X}_k^{x,-} = \begin{bmatrix} 0.5 & 1.1 & 0.7 & 0.4 & 1.0 \\ 0.6 & 0.2 & 0.2 & 0.7 & 0.8 \end{bmatrix}$$

and that $\alpha_k^{(c)} = 0.2$ for all k . We can compute that

$$\Sigma_{\bar{x},k}^- = \begin{bmatrix} 0.0744 & -0.0220 \\ -0.0220 & b \end{bmatrix}.$$

What is b ? Enter your answer with three digits to the right of the decimal point.

0.064

✔

Correct

Yes. That is the correct answer.